

Aspectual Classification Revisited

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1 Aspectual Classification

One of the key insights in lexical semantic classification of verbs is that verbs can be distinguished based on how the events they denote take place in time. According to Dowty (1979), such classification originated with Aristotle in *Metaphysics* and was later refined by Oxford philosophers such as Gilbert Ryle and Anthony Kenny. Then Vendler (1967) proposed a four-way distinction that is widely adopted in the current literature: STATES, ACTIVITIES, ACCOMPLISHMENTS, and ACHIEVEMENTS. Examples of verbs from Vendler's four categories, drawn from Dowty (1979), are listed below:

STATES	ACTIVITIES	ACCOMPLISHMENTS	ACHIEVEMENTS
know	run	paint a picture	recognize
believe	walk	make a chair	spot
have	swim	deliver a sermon	find
desire	push a cart	draw a circle	lose
love	drive a car	push a cart	reach
		recover from illness	die

Dowty (1979) explores a variety of diagnostics for distinguishing between these verb classes.¹ For example, he introduces the progressive test and the pseudo-cleft test, among others, as tools to differentiate STATES from ACTIVITIES and ACCOMPLISHMENTS.

- | | | | |
|-----|----|---|----------------|
| (1) | a. | *John is knowing the answer. ² | STATE |
| | b. | John is running. | ACTIVITY |
| | c. | John is building a house. | ACCOMPLISHMENT |
| (2) | a. | *What John did was know the answer. | |
| | b. | What John did was run. | |
| | c. | What John did was build a house. | |

¹He does not explicitly use the term *diagnostics*, though.

²In this paper, I use asterisks (*) to indicate unacceptability, whereas the original texts may use hash signs (#) for the same purpose.

He also highlights Kenny's observation regarding differences between STATES and other verb classes in the default interpretation of the simple present tense:

As Kenny noted, when an activity or accomplishment occurs in the simple present tense (or in any non-progressive tense), it has a frequentative (or habitual) interpretation in normal contexts. If (3b) and (3c) are not used in one of a few specialized contexts (e.g. used by an announcer at a sports event, appear as a stage direction, appear in a narrative in the historical present), then they are understood to involve more than one event of reciting a poem or running respectively. But (3a) does not involve more than one occasion of knowing the answer. (The third example is changed from *build a house* to *recite a poem*, because one cannot build the same house more than once, so the frequentative interpretation would be problematic.) [In the original text, the examples are presented as (26a-c).]

- (3) a. John knows the answer.
- b. John runs.
- c. John recites a poem.

Furthermore, Dowty shows that ACTIVITIES and ACCOMPLISHMENTS exhibit different behaviors concerning adverbial prepositional phrases and entailments from the progressive to the non-progressive tenses.

- (4) a. ?John painted a picture for an hour. ACCOMPLISHMENT
- b. John painted a picture in an hour.
- (5) a. John walked for an hour. ACTIVITY
- b. *John walked in an hour.
- (6) a. John is painting a picture. $\nRightarrow$ John has painted a picture.
- b. John is running. $\Rightarrow$ John has run.

Dowty adds to this series of tests the observation that only ACCOMPLISHMENT verbs can normally occur as the complement of *finish*:

- (7) a. John finished painting a picture.
- b. *John finished walking.

Finally, Dowty points out as characteristics of ACHIEVEMENT verbs that they are (i) generally quite strange with a *for*-phrase, (ii) generally unacceptable as complements of *finish*, and (iii) unlike both ACCOMPLISHMENTS and ACTIVITIES, unacceptable as complements of *stop* (except in a habitual reading).

- (8) a. John noticed the painting in a few minutes.
- b. ??John noticed the painting for a few minutes.

- (9) *John finished noticing the painting.
 (10) *John stopped noticing the painting.

Modern approaches to the aspectual classification of verbs have identified a fifth category: SEMELFACTIVES. Some scholars have suggested that SEMELFACTIVES can be grouped with ACHIEVEMENTS as point events because of their typical brevity. However, unlike ACHIEVEMENTS, SEMELFACTIVES are not generally modifiable with *in* adverbials (Kearns, 2010).

- (11) a. *Jones rapped the table in a moment.
 b. *Jones blinked in a minute.
 c. *Jones coughed in an hour.
 d. *The light flashed in ten minutes.

Another widely accepted view in the current literature is that different aspectual classes can be characterized using notions such as *dynamicity*, *durativity*, and *telicity*. For instance, Olsen (2016) proposes a feature-based analysis grounded in the three aspectual notions. According to Olsen’s analysis, both SEMELFACTIVES and ACHIEVEMENTS are dynamic and non-durative. The key distinction between the two is that SEMELFACTIVES are atelic, while ACHIEVEMENTS are telic.³

(12)	Dynamic	Durative	Telic	
Accomplishment	+	+	+	<i>draw, build</i>
Activity	+	+	0	<i>run, chat, play, gurgle</i>
Achievement	+	0	+	<i>notice, recognize, win</i>
Semelfactive	+	0	0	<i>jump, hit, tap, cough</i>
Temporary State	0	+	+	<i>be sick/available</i>
(Permanent) State	0	+	0	<i>be tall, know, contain</i>
[UNATTESTED]	0	0	+	
[UNATTESTED]	0	0	0	

Despite the progress made in modern approaches to aspectual classification, there remains ongoing controversy surrounding the reliability of diagnostics. Many have been challenged, and no comprehensive list of diagnostics has yet been established that consistently yields reliable results for aspectual classification. Problems are noted by Dowty (1979) himself; for example, some stative verbs pass the progressive test, contrary to prior discussions. Accordingly, Dowty acknowledges that STATES may not constitute a homogeneous class, a concept reminiscent of the distinction between individual-level and stage-level predicates.

³Olsen (2016) in fact argues that aspectual features are privative, meaning that verb classes marked with zero (0) are unspecified for that aspectual notion. For example, ACHIEVEMENTS are dynamic and telic, but unspecified for *durativity*, meaning they can be either durative or non-durative. I believe that this distinction is not crucial to the analysis to be developed in this paper, and will assume that verb classes are fully specified for each aspectual notion.

- (13) a. The socks are lying under the bed.
b. Your glass is sitting near the edge of the table.

Some diagnostics have been shown to be affected by non-aspectual factors like agentivity (Kearns, 2010). Among the diagnostics previously discussed, the *x finished ϕ -ing* test is an example that falls into this category.

- (14) a. John finished painting a picture.
b. ??John finished recovering from illness.

A summary of diagnostics and their application to the following examples from each verb class is presented in (22). (See section A for more details.)

- (15) INDIVIDUAL-LEVEL STATES
Pat is altruistic. / Tony knows French.
- (16) STAGE-LEVEL STATES
The machine was noisy. / The keys lay on the bookcase.
- (17) SEMELFACTIVES
Sam hopped. / The timer beeped.
- (18) ACTIVITIES
Tracy ran. / Devon pushed a cart. / The wheel spun.
- (19) ACCOMPLISHMENTS
Kim drew a circle. / Casey read a paper.
- (20) ASSOCIATED-PROCESS ACHIEVEMENTS
The train arrived at the station. / The climber reached the summit.
- (21) ‘LUCKY’ ACHIEVEMENTS
Kelly won the lottery. / The appraiser noticed a flaw in the diamond.
- (22) *Summary of Diagnostics*

Diagnostic	States		Semelfactives	Activities
	Individ. lev.	Stage lev.		
present tense is present	○	×	×	×
<i>what happened was ϕ</i>	×	×	○	○
progressive	×	○	○-iter	○
<i>ϕ for an hour</i> (‘during’)	×	○	○-iter	○
<i>ϕ in an hour</i> (‘within’)	×	×	×	×
<i>x is ϕ-ing entails x has ϕ-ed</i>	N/A	○	○-iter	○
<i>x stopped ϕ-ing</i>	N/A	○?	○	○
<i>x finished ϕ-ing</i>	N/A	×	○	?

Diagnostic	Accomplishments	Achievements	
		Assoc. proc.	‘Lucky’
present tense is present	×	×	×
<i>what happened was ϕ</i>	○	○	○
progressive	○	○	×
<i>ϕ for an hour</i> (‘during’)	○	×	×
<i>ϕ in an hour</i> (‘within’)	○	○	×
<i>x is ϕ-ing entails x has ϕ-ed</i>	×	×	N/A
<i>x stopped ϕ-ing</i>	○	?	N/A
<i>x finished ϕ-ing</i>	○	×	N/A

That said, the table in (22) does not provide a clear representation of how diagnostics apply to aspectual classification. This observation underscores the central aim of this study: to revisit aspectual classification and its associated diagnostics, and to explore whether refining certain details or eliminating specific components can lead to a better understanding.

The rest of the paper is organized as follows. In section 2, I argue that dynamicity, or the distinction between statives and non-statives, is in fact not an aspectual notion. In section 3, I reassess the diagnostics outlined in Dowty (1979) and develop a comprehensive list of diagnostics based on a proposed criterion. Once dynamicity, and accordingly STATES, are excluded from the picture, I demonstrate that the selected diagnostics yield reliable results for aspectual classification. In section 4, I address associated-process ACHIEVEMENT verbs, which present critical empirical challenges to the proposed diagnostics. In section 5, I revisit Olsen’s (2016) feature-based analysis, proposing a minor revision regarding the distinction between individual-level and stage-level STATES. Finally, section 6 summarizes the findings and conclusions.

2 Dynamicity and States

The driving question of this study is the following: Is dynamicity truly an aspectual notion? In other words, does it determine or imply anything about how an eventuality is structured within time? On conceptual grounds, there seems to be sufficient reason to believe it does not, as demonstrated in the following visual representations:

(23) *Visual Representations of Aspectual Classes*

- a. ACCOMPLISHMENTS: —————●
- b. ACTIVITIES: —————○
- c. ACHIEVEMENTS: ●
- d. SEMELFACTIVES: ○
- e. STATES: ?

While the specifics may vary, a common approach to aspectual classification involves representing a verb’s temporal structure in terms of intervals (Dowty, 1979; Kearns, 2010; Olsen, 2016). Telicity refers to the property of having a culmination, telos, or endpoint. This distinction can be visually represented by a closed dot (telic) versus an open dot (atelic) at the end of the interval. A durative event extends over a period of time, in contrast to a non-durative event, which is idealized as occurring at a single point in time. This distinction can be visually represented as a line (durative) versus a dot (non-durative).

Together, durativity and telicity form a fully representative four-way distinction: ACCOMPLISHMENTS are telic and durative, ACTIVITIES are atelic

and durative, ACHIEVEMENTS are telic and non-durative, and SEMELFACTIVES are atelic and non-durative. However, dynamicity does not align in the same way with durativity and telicity. As Olsen (2016) points out, some verb classes predicted to exist under a feature-based approach are, in fact, unattested; see (12). Some researchers, including Dowty (1979) and Olsen (2016), suggest that STATES are heterogeneous with respect to aspect. Yet even in such theories, dynamicity does not interact freely with other aspectual notions to produce a full eight-way distinction.

Dynamicity distinguishes STATES from other verb classes, collectively referred to as events, with the key difference being whether an eventuality involves change. I believe most would agree that this notion is somewhat distinct from telicity and durativity; it is difficult to conceptualize a way to visually represent static versus dynamic in the same manner as in (23). Therefore, it seems reasonable to exclude dynamicity from the set of aspectual notions. Although dynamicity — and the distinction between STATES and events — may prove useful in explaining various linguistic phenomena, I contend that it bears no direct relevance to aspect.

But is there any empirical basis for excluding dynamicity and STATES altogether? While I do not currently observe direct evidence, I propose that indirect evidence may justify such an approach. By revisiting the aspectual diagnostics outlined by Dowty (1979), I demonstrate that the overall framework becomes clearer when STATES are excluded from the aspectual classes of verbs.

3 Reassessing Diagnostics

As noted in section 1, several diagnostics introduced by Dowty (1979) have been questioned by many scholars including Dowty himself. Therefore, a good starting point for reassessment would be to set up a criterion based on which diagnostics can be carefully selected. I propose the following aspectual classification diagnostic criterion⁴:

(24) *Aspectual Classification Diagnostic Criterion*

Diagnostics used in the aspectual classification of verbs must target semantically well-established aspectual notions — durativity and telicity — and nothing else.

This criterion allows us to validate diagnostics on purely semantic grounds, prior to evaluating their application and empirical outcomes. It also helps eliminate problematic diagnostics, such as the *x finished ϕ -ing* test, which has been shown to be influenced by non-aspectual factors like agentivity. Based on this criterion, I focus on four diagnostics that hold particular

⁴A similar idea has been proposed by Olsen (2016), who wrote, “Although the frames are sometimes informally described as tests for class membership, they are actually sensitive to individual features distinguishing the classes from each other.”

value: (i) progressive and *ϕ for an hour* as durativity tests, (ii) *ϕ in an hour* as a telicity test, and (iii) *x is ϕ-ing entails x has ϕ-ed* as a telicity test applicable only to durative verbs.

First, the progressive test and the *ϕ for an hour* test target durativity. For an event to be in progress or extend over some significant time period, it must be durative. Therefore, we expect ACTIVITIES and ACCOMPLISHMENTS to pass these tests, while SEMELFACTIVES and ACHIEVEMENTS fail. This prediction is (partially) borne out:

- (25) a. Sam was hopping. / The timer was beeping.
- b. Tracy was running. / Devon was pushing a cart. / The wheel was spinning.
- c. Kim was drawing a circle. / Casey was reading a paper.
- d. ?The train was arriving at the station. / The climber was reaching the summit.
- e. *Kelly was winning the lottery. / *The appraiser was noticing a flaw in the diamond.
- (26) a. Sam hopped for an hour. / The timer beeped for an hour.
- b. Tracy ran for an hour. / Devon pushed a cart for an hour. / The wheel spun for an hour.
- c. Kim drew a circle for an hour. / Casey read a paper for an hour.
- d. *The train arrived at the station for an hour. / *The climber reached the summit for an hour.
- e. *Kelly won the lottery for an hour. / *The appraiser noticed a flaw in the diamond for an hour.

SEMELFACTIVES are compatible with both durativity tests, but yield iterative interpretations. For instance, the progressive forms in (25a) do not indicate that the event denoted by the semelfactive verbs extend over a time interval; rather, it suggests that the event occurs iteratively. Therefore, when applying the durativity tests, it is essential to consider not only the acceptability of a sentence but also its interpretation. Additionally, associated-process ACHIEVEMENT verbs present another challenge: they are compatible with progressives but not with *for*-adverbials. We return to this issue in the next section.

Turning to telicity tests, the *ϕ in an hour* test specifically targets telicity. Telic predicates have a culmination, telos, or endpoint, that allows them to occur with time adverbial phrases such as *in an hour*. Therefore, we expect ACHIEVEMENTS and ACCOMPLISHMENTS to pass the *ϕ in an hour* test, while SEMELFACTIVES and ACTIVITIES fail. This prediction is (again, partially) borne out:

- (27) a. *Sam hopped in an hour. / *The timer beeped in an hour.
 b. *Tracy ran in an hour. / *Devon pushed a cart in an hour. /
 *The wheel spun in an hour.
 c. ?Kim drew a circle in an hour. / Casey read a paper in an hour.
 d. ?The train arrived at the station in an hour. / The climber
 reached the summit in an hour.
 e. *Kelly won the lottery in an hour. / ??The appraiser noticed a
 flaw in the diamond in an hour.

Example sentences with a question mark, such as (27c-1) and (27e-2), may suggest that *in*-adverbials are simply incompatible with these constructions. However, the questionable status of these sentences is not due to the aspectual properties of the predicates but rather to the temporal specification of the adverbial, i.e. *an hour*. Based on our world knowledge, events in which *the appraiser notices a flaw in the diamond* or *Kim draws a circle* hardly ever occur over a timespan of *an hour*. As such, an alternative time adverbial A SECOND resolves the issue, as it aligns better with our perceptions of the temporal properties of the events described in the examples.

- (28) a. Kim drew a circle in an hour.
 b. The appraiser noticed a flaw in the diamond in a second.

Finally, the *x is ϕ -ing entails x has ϕ -ed* test targets telicity, but is only applicable to durative verbs, because as discussed earlier, non-durative verbs are incompatible with progressives. The implication of this diagnostic is very much related to Kearns's (2010) discussion of the *sub-interval property*.

- (29) a. Jones ate a sandwich. ACCOMPLISHMENT
 b. Jones watched TV. ACTIVITY

Let us denote the interval during which the event described by (29a) occurs as I_α . Now consider a sub-interval I_β : while this sub-interval also contains an event, it does not encompass the event *Jones ate a sandwich*. Instead, I_β represents the eating process leading up to, but not including, the culmination point of the event. In contrast, the description *Jones watched TV* accurately captures the contents of the sub-interval I_β , as well as any other sub-intervals of the event's run time. This distinction generalizes as follows: atelic events exhibit the sub-interval property, whereas telic events do not.

Under the assumption that progressives denote a sub-interval of an event (Dowty, 1979), we predict that *x is ϕ -ing* entails *x is ϕ -ed* in the case of ACTIVITIES, but not for ACCOMPLISHMENTS. This prediction is borne out:

- (30) a. Tracy is running. $\Rightarrow$ Tracy has run.
 Devon is pushing a cart. $\Rightarrow$ Devon has pushed a cart.
 The wheel is spinning. $\Rightarrow$ The wheel has spun.
 b. Kim is drawing a circle. $\nRightarrow$ Kim has drawn a circle.
 Casey is reading a paper. $\nRightarrow$ Casey has read a paper.

To summarize, this section introduced a curated set of diagnostics drawn from Dowty (1979), chosen based on a criterion ensuring both their semantic foundation and empirical robustness. A summary of these selected diagnostics is presented below:

(31) *Summary of Selected Diagnostics*

Diagnostic	Semelfactives	Activities
progressive	○-iter	○
ϕ for an hour ('during')	○-iter	○
ϕ in an hour ('within')	×	×
x is ϕ -ing entails x has ϕ -ed	○-iter	○

Diagnostic	Accomplishments	Achievements	
		Assoc. proc.	'Lucky'
progressive	○	○	×
ϕ for an hour ('during')	○	×	×
ϕ in an hour ('within')	○	○	×
x is ϕ -ing entails x has ϕ -ed	×	×	N/A

Before proceeding to the next section, a few notes are warranted. First, as emphasized earlier, it is crucial to evaluate these tests not only based on sentence acceptability but also on interpretation. Specifically, an iterative interpretation of SEMELFACTIVES should be treated as a negative result, indicating their non-durative nature.

Second, the only gaps in the empirical landscape of (31) pertain to ACHIEVEMENTS. One subset of ACHIEVEMENT verbs, known as associated-process ACHIEVEMENTS, is unexpectedly compatible with progressives. Another subset, referred to as "lucky" ACHIEVEMENTS, appears generally incompatible with the adverbial phrase *in an hour*. The next section focuses on addressing this issue.

4 Associated-Process Achievement Verbs

It has long been observed that certain examples of ACHIEVEMENT verbs exhibit behavior similar to ACCOMPLISHMENTS. For instance, Kearns (2010), discussing what he terms the *prelude effect*, notes that "*win, die, reach the summit, and arrive* describe the culminations of events which have the structure of accomplishments." These verb phrases resemble ACCOMPLISHMENTS in that they can occur in progressive forms and are generally more compatible with extended time adverbials.

From the perspective of visual representations, a range of telic predicates can be illustrated as follows:

(32) *Visual Representations of Telic Predicates*

- a. ACCOMPLISHMENTS: —————●
- b. Associated-process ACHIEVEMENTS:●
- c. Canonical ACHIEVEMENTS: ●

I propose that associated-process ACHIEVEMENTS, due to the presence of associated processes, readily shift between aspectual classes, alternating between ACHIEVEMENTS and ACCOMPLISHMENTS. Aspectual class shifts along the durativity axis are also observed in atelic predicates. The use of *burp* in its progressive form and with durative adverbials are ambiguous between an iterative interpretation and an ‘extended’ interpretation (e.g. in a world burping championship scenario):

(33) He has been burping for over 10 seconds!

The ambiguity of (33) illustrates a shift in aspectual class between SEMELFACTIVES and ACTIVITIES. Aspectual class shifts along the durativity axis are likely permissible because the duration of an event is inherently a continuous concept rather than a discrete one.

5 Aspectual Classification within States

Although I argued in section 2 for excluding STATES from aspectual classification, this does not imply that the verb class identified as STATES resists analysis in terms of aspect. As noted earlier, some scholars have observed that STATES constitute a non-homogeneous class in aspectual classification. Building on the diagnostics established in section 3, I revisit the aspectual classification of STATES.

First, stage-level STATES, such as *be noisy* or *lie*, appear to be more compatible with durativity tests compared to individual-level STATES, such as *be altruistic* or *know*. In fact, *be altruistic* is ambiguous between individual-level and stage-level readings. However, coercion of *be altruistic* into a stage-level reading is observed only with the progressive test and not with the *ϕ for an hour* test. This observation might suggest that the *ϕ for an hour* test is more reliable than the progressive test for probing durativity. The surprising compatibility of associated-process ACHIEVEMENTS with the progressive form could therefore warrant an alternative analysis, though I will not explore this further in the present paper.

- (34) a. ?Pat is being altruistic. / *Tony is knowing French.
b. ?The machine was being noisy. / The keys were lying on the bookcase.
- (35) a. *Pat is altruistic for an hour. / *Tony knows French for an hour.
b. The machine was noisy for an hour. / The keys lay on the bookcase for an hour.

Telicity tests suggest that STATES, both individual-level and stage-level, are atelic. They are incompatible with expressions like *in an hour* and exhibit entailment relationships between *x is ϕ -ing* and *x has ϕ -ed* (the former entails the latter).

- (36) a. *Pat is altruistic in an hour. / *Tony knows French in an hour.
 b. *The machine was noisy in an hour. / *The keys lay on the
 bookcase in an hour.
- (37) a. ?Pat is being altruistic. $\Rightarrow$ Pat has been altruistic.
 b. ?The machine is being noisy. $\Rightarrow$ The machine has been noisy.
 c. The keys are lying on the bookcase. $\Rightarrow$?The keys have lain on
 the bookcase.

In summary, here are the results of applying the selected diagnostics to individual-level and stage-level STATES:

- (38) *Summary of Selected Diagnostics* (STATES only)

Diagnostic	States	
	Individ. lev.	Stage lev.
progressive	Δ	$\bigcirc$
ϕ for an hour ('during')	$\times$	$\bigcirc$
ϕ in an hour ('within')	$\times$	$\times$
<i>x is ϕ-ing entails x has ϕ-ed</i>	N/A	$\bigcirc$

The results indicate one thing: STATES are all atelic, the distinction between individual-level and stage-level possibly being non-durative vs. durative. If this analysis is correct, it may provide deeper insight into the concept of durativity. Durativity does not simply distinguish between *punctual* and *durative*, but rather (in interval semantics terms) pertains to whether a discernible time interval can be identified for a given eventuality. Individual-level STATES, being an inherent property to some entity or individual, cannot be identified with a discernible time interval, which explains their non-durative nature.

STATES are thus privative eventualities in that they inherently lack a natural endpoint or telos. In fact, similar view have been alluded to by previous researchers, as cited in Olsen (2016): According to Passoneau (1988:47), because states do not denote change (KINESIS in her model), “they cannot culminate in new situations,” i.e. be telic. According to Tenny (1994), states may not be “delimited.”

Nevertheless, Olsen (2016) analyzes stage-level STATES as [+telic, +durative] and individual-level STATES as [+durative]. I propose a minor revision: the distinction between individual-level and stage-level STATES should be understood in terms of durativity, not telicity.

6 Conclusion

In this paper, I revisited the aspectual classification of verbs and the diagnostics proposed by Dowty (1979). I first argued that dynamicity is not an aspectual notion and, therefore, should be excluded from the aspectual classification system. Next, I demonstrated that the selected diagnostics grounded in a specific criterion, coupled with the exclusion of STATES, offer a clearer framework that improves our understanding of aspectual classification. Two key implications emerged from the empirical results of applying these diagnostics to various verb classes: First, associated-process ACHIEVEMENTS were shown to undergo shifts in aspectual class. Second, I argued that STATE are inherently atelic, with the distinction between individual-level and stage-level STATES being one of durativity, which, though counterintuitive, is consistent with the data.

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Appendix

A Summary of Diagnostics

- (1) present tense is present
- a. Pat is altruistic. / Tony knows French.
 - b. The machine is noisy. / The keys lie on the bookcase.
 - c. Sam hops. / The timer beeps.
 - d. Tracy runs. / Devon pushes a cart. / The wheel spins.
 - e. Kim draws a circle. / Casey reads a paper.
 - f. The train arrives at the station. / The climber reaches the summit.
 - g. Kelly wins the lottery. / The appraiser notices a flaw in the diamond.

Frequentative (or habitual) interpretation for (1b–g) in normal contexts.

- (2) *what happened was ϕ*
- a. *What happened was Pat was altruistic. / *What happened was Tony knew French.
 - b. ??What happened was the machine was noisy. / *What happened was the keys lay on the bookcase.
 - c. What happened was Sam hopped. / What happened was the timer beeped.
 - d. What happened was Tracy ran. / What happened was Devon pushed a cart. / What happened was the wheel spun.
 - e. What happened was Kim drew a circle. / What happened was Casey read a paper.
 - f. What happened was the train arrived at the station. / What happened was the climber reached the summit.
 - g. What happened was Kelly won the lottery. / What happened was the appraiser noticed a flaw in the diamond.

- (3) progressive
- a. ?Pat is being altruistic. / *Tony is knowing French.
 - b. ?The machine was being noisy. / The keys were lying on the bookcase.
 - c. Sam was hopping. / The timer was beeping.
 - d. Tracy was running. / Devon was pushing a cart. / The wheel was spinning.
 - e. Kim was drawing a circle. / Casey was reading a paper.
 - f. ?The train was arriving at the station. / The climber was reaching the summit.
 - g. *Kelly was winning the lottery. / *The appraiser was noticing a flaw in the diamond.

SEMELFACTIVES in the progressive form (3c) yield iterative interpretations.

- (4) ϕ *for an hour* ('during')
- a. *Pat is altruistic for an hour. / *Tony knows French for an hour.
 - b. The machine was noisy for an hour. / The keys lay on the bookcase for an hour.
 - c. Sam hopped for an hour. / The timer beeped for an hour.
 - d. Tracy ran for an hour. / Devon pushed a cart for an hour. / The wheel spun for an hour.
 - e. Kim drew a circle for an hour. / Casey read a paper for an hour.
 - f. *The train arrived at the station for an hour. / *The climber reached the summit for an hour.
 - g. *Kelly won the lottery for an hour. / *The appraiser noticed a flaw in the diamond for an hour.

SEMELFACTIVES with the adverbial phrase *for an hour* also lead to iterative interpretations.

- (5) ϕ *in an hour* ('within')
- a. *Pat is altruistic in an hour. / *Tony knows French in an hour.
 - b. *The machine was noisy in an hour. / *The keys lay on the bookcase in an hour.
 - c. *Sam hopped in an hour. / *The timer beeped in an hour.
 - d. *Tracy ran in an hour. / *Devon pushed a cart in an hour. / *The wheel spun in an hour.
 - e. ?Kim drew a circle in an hour. / Casey read a paper in an hour.
 - f. ?The train arrived at the station in an hour. / The climber reached the summit in an hour.
 - g. *Kelly won the lottery in an hour. / ??The appraiser noticed a flaw in the diamond in an hour.

Some examples improve when paired with an alternative *for*-phrase with a brief timespan.

- (5') a. Kim drew a circle in five seconds.
b. The appraiser noticed a flaw in the diamond in a second.
- (6) x is ϕ -ing entails x has ϕ -ed test (not applicable if *progressive)
- a. ?The machine is being noisy. $\Rightarrow$ The machine has been noisy.
The keys are lying on the bookcase. $\Rightarrow$?The keys have lain on the bookcase.
 - b. Sam is hopping. $\Rightarrow$ Sam has hopped.
The timer is beeping. $\Rightarrow$ The timer has beeped.
 - c. Tracy is running. $\Rightarrow$ Tracy has run.
Devon is pushing a cart. $\Rightarrow$ Devon has pushed a cart.
The wheel is spinning. $\Rightarrow$ The wheel has spun.

- d. Kim is drawing a circle. $\nRightarrow$ Kim has drawn a circle.
Casey is reading a paper. $\nRightarrow$ Casey has read a paper.
- e. The train is arriving at the station. $\nRightarrow$ The train has arrived at the station.
The climber is reaching the summit. $\nRightarrow$ The climber has reached the summit.

For SEMELFACTIVES, the entailment relationships in (6b) hold when interpreted iteratively.

- (7) *x stopped ϕ -ing* (not applicable if *progressive)
 - a. The machine stopped being noisy. / ??The keys stopped lying on the bookcase.
 - b. Sam stopped hopping. / The timer stopped beeping.
 - c. Tracy stopped running. / Devon stopped pushing a cart. / The wheel stopped spinning.
 - d. Kim stopped drawing a circle. / Casey stopped reading a paper.
 - e. ??The train stopped arriving at the station. / ??The climber stopped reaching the summit.
- (8) *x finished ϕ -ing* (not applicable if *progressive)
 - a. *The machine finished being noisy. / *The keys finished lying on the bookcase.
 - b. Sam finished hopping. / ?The timer finished beeping.
 - c. Tracy finished running. / ?Devon finished pushing a cart. / *The wheel finished spinning.
 - d. Kim finished drawing a circle. / Casey finished reading a paper.
 - e. *The train finished arriving at the station. / ??The climber finished reaching the summit.